SIMATS ENGINEERING

CO3- AT2 - Design Task - Medium

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 3 - *Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems.* (BL6)

Assessment Tool Weightage: 35%

Short description about assessment: Students design a system / model based on given requirements to assess creativity and application skills.

Dr.

QUESTIONS: 1 (1 X 100 = 100) **Total Marks: 100 Duration: 3hrs**

1. Design a system architecture for Multi-access Edge Computing to reduce latency in smart traffic applications. (BL6)
2. Develop a cloud-edge communication model for wireless healthcare monitoring. (BL6)
3. Design a functional architecture for real-time edge data processing in industrial automation. (BL6)
4. Propose a cloud-based wireless framework for smart surveillance systems. (BL6)
5. Design an edge AI workflow for intelligent wireless data filtering. (BL6)
6. Develop a task offloading strategy between edge and cloud in 5G applications. (BL6)
7. Design a low-latency communication architecture for edge-assisted smart city services. (BL6)
8. Propose a scalable edge-cloud framework for wireless sensor networks. (BL6)
9. Design a Cloud Radio Access Network model for dense urban deployment. (BL6)
10. Develop a functional split model between centralized and distributed radio units. (BL6)
11. Design a virtualized wireless access framework for next-generation communication. (BL6)
12. Propose a data flow model for cloud-based radio resource allocation. (BL6)
13. Design a centralized traffic control model using Cloud RAN principles. (BL6)
14. Develop a network management architecture for cloud-assisted wireless systems. (BL6)

15. Design a wireless service framework integrating virtual network functions. (BL6)
16. Propose an optimized fronthaul communication model. (BL6)
17. Design a satellite-assisted communication model using Low Earth Orbit satellite for remote wireless coverage. (BL6)
18. Develop a comparative architecture for LEO, MEO, and GEO backhaul systems. (BL6)
19. Design a spectrum sharing model between LTE and 5G users. (BL6)
20. Propose a frequency allocation strategy for mixed wireless services. (BL6)
21. Design a dynamic spectrum access framework for urban wireless systems. (BL6)
22. Develop a communication architecture combining terrestrial and satellite networks. (BL6)
23. **Design a spectrum efficiency improvement model for dense communication zones. (BL6)**
24. Propose a channel reuse plan for multi-user wireless communication. (BL6)
25. Design an NB-IoT architecture for smart utility monitoring. (BL6)
26. Develop an LTE-M communication workflow for wearable monitoring systems. (BL6)
27. Design a low-power communication model for battery-operated IoT devices. (BL6)
28. Propose a device communication plan for long-life sensor networks. (BL6)
29. Design a 5G-enabled IoT framework for campus automation. (BL6)
30. Develop an energy-efficient wireless communication strategy for sensor nodes. (BL6)
31. Design an IoT communication flow for periodic environmental sensing. (BL6)
32. Propose a scalable wireless architecture for smart connected devices. (BL6)
33. Design a smart city communication framework for traffic monitoring and surveillance. (BL6)
34. Develop an energy monitoring communication model for public infrastructure. (BL6)
35. Design a smart factory information flow for predictive maintenance. (BL6)
36. Propose a robotic automation communication framework for industrial control. (BL6)
37. Design a remote patient monitoring communication architecture. (BL6)
38. Develop a wireless communication workflow for AI-assisted healthcare diagnosis. (BL6)
39. Design a smart irrigation communication model for crop monitoring. (BL6)
40. Propose a livestock tracking communication architecture using wireless sensors. (BL6)

Code of Conduct:

I A P Navya Shri(192312028) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

A P Navya Shri

Signature of the student:

RUBRICS FOR CALCULATION:

Design Task					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Design	20%	Demonstrates deep understanding of design objectives, constraints, and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Lacks understanding of the design context
Evaluation of Design	25%	Provides in-depth critique identifying strengths, weaknesses, and design gaps	Identifies key issues with reasonable analysis	Limited critique; mostly descriptive feedback	No meaningful critique provided
Justification	20%	Supports critique with strong reasoning, evidence, and relevant principles	Provides justification with some supporting evidence	Weak or unclear justification	No justification for feedback
Suggestions for Improvement	20%	Offers innovative, practical, and well-justified improvement suggestions	Provides useful suggestions with minor gaps	Suggestions are limited or partially relevant	No constructive suggestions provided
Communication	15%	Communicates feedback clearly, professionally, and engages actively in discussion	Communicates well with minor issues	Communication lacks clarity or engagement	Poor communication and minimal participation

Q. Design a spectrum efficiency improvement model for dense communication zones.

Task

To design a spectrum efficiency improvement model for dense communication zones using adaptive transmit power control and interference management, and compare its performance with a conventional communication system.

Description

In dense communication zones, many users and base stations operate within a limited area. This causes high co-channel interference and reduces spectrum efficiency.

The proposed model uses adaptive power control to reduce unnecessary interference. The system calculates the SINR (Signal-to-Interference-plus-Noise Ratio) and spectrum efficiency for both conventional and proposed systems.

Spectrum efficiency is calculated using:

$SE = \log_2(1+SINR)$

The performance is compared based on SINR, spectrum efficiency, throughput, and transmit power.

Parameters Used

Parameter	Value
Simulation Area	500 × 500 m
Number of Base Stations	10
Users per Base Station	10
Total Users	100
Carrier Frequency	2.4 GHz
Bandwidth	20 MHz
Maximum Transmit Power	1 W
Minimum Transmit Power	0.1 W
Path Loss Exponent	3.5
Reference Path Loss	40 dB
Noise Power Density	-174 dBm/Hz

Noise Figure	9 dB
Target SINR	10 dB

Code:

```
clc;
clear;
close all;

rng(10);

% Parameters
areaSize = 500;
numBS = 10;
usersPerBS = 10;
numUsers = numBS * usersPerBS;

BW = 20e6;
Pt_max = 1;
Pt_min = 0.1;

pathLossExp = 3.5;
PL0 = 40;

noise_dBm_Hz = -174;
noiseFigure = 9;

% Noise power
noise_dBm = noise_dBm_Hz + 10*log10(BW) + noiseFigure;
noise_W = 10^((noise_dBm - 30)/10);

% Base station locations
BS = areaSize * rand(numBS,2);

% User locations
```

```
users = areaSize * rand(numUsers,2);

% Find serving base station
servingBS = zeros(numUsers,1);

for u = 1:numUsers
    distances = sqrt(sum((BS - users(u,:)).^2,2));
    [~, servingBS(u)] = min(distances);
end

% Distance between users and base stations
distance = zeros(numUsers,numBS);

for u = 1:numUsers
    for b = 1:numBS
        distance(u,b) = sqrt(sum((users(u,:) - BS(b,:)).^2));
    end
end

distance(distance < 1) = 1;

% Path loss
pathLoss = PL0 + 10*pathLossExp*log10(distance);

% Channel gain
gain = 10.^(-pathLoss/10);

%% Conventional System

Pt_conv = Pt_max * ones(numBS,1);

SINR_conv = zeros(numUsers,1);
SE_conv = zeros(numUsers,1);

for u = 1:numUsers
```

```
    b = servingBS(u);

    signal = Pt_conv(b) * gain(u,b);

    interference = 0;

    for k = 1:numBS
        if k ~= b
            interference = interference + ...
                Pt_conv(k) * gain(u,k);
        end
    end

    SINR_conv(u) = signal / (interference + noise_W);
    SE_conv(u) = log2(1 + SINR_conv(u));

end

%% Proposed Adaptive Power Control

Pt_prop = Pt_max * ones(numBS,1);

targetSINR = 10^(10/10);

for iteration = 1:10

    for b = 1:numBS

        userIndex = find(servingBS == b);

        if isempty(userIndex)
            continue;
        end
```

```
        avgSINR = 0;

        for u = userIndex'

            signal = Pt_prop(b) * gain(u,b);

            interference = 0;

            for k = 1:numBS
                if k ~= b
                    interference = interference + ...
                        Pt_prop(k) * gain(u,k);
                end
            end

            currentSINR = signal / ...
                (interference + noise_W);

            avgSINR = avgSINR + currentSINR;

        end

        avgSINR = avgSINR / length(userIndex);

        if avgSINR < targetSINR
            Pt_prop(b) = min(Pt_prop(b)*1.10, Pt_max);
        else
            Pt_prop(b) = max(Pt_prop(b)*0.90, Pt_min);
        end

    end
end

%% Proposed SINR and Spectrum Efficiency
```

```
SINR_prop = zeros(numUsers,1);
SE_prop = zeros(numUsers,1);

for u = 1:numUsers

    b = servingBS(u);

    signal = Pt_prop(b) * gain(u,b);

    interference = 0;

    for k = 1:numBS
        if k ~= b
            interference = interference + ...
                Pt_prop(k) * gain(u,k);
        end
    end

    SINR_prop(u) = signal / (interference + noise_W);
    SE_prop(u) = log2(1 + SINR_prop(u));

end

% Convert SINR to dB
SINR_conv_dB = 10*log10(SINR_conv);
SINR_prop_dB = 10*log10(SINR_prop);

% Average spectrum efficiency
avgSE_conv = mean(SE_conv);
avgSE_prop = mean(SE_prop);

% Throughput
throughput_conv = BW * sum(SE_conv);
throughput_prop = BW * sum(SE_prop);
```

```
% Average power
avgPower_conv = mean(Pt_conv);
avgPower_prop = mean(Pt_prop);

% Improvement
SE_improvement = ...
    ((avgSE_prop - avgSE_conv)/avgSE_conv)*100;

%% Display Output

fprintf('\nSPECTRUM EFFICIENCY ANALYSIS\n');
fprintf('-----------------------------\n');

fprintf('Number of Base Stations : %d\n',numBS);
fprintf('Number of Users : %d\n',numUsers);

fprintf('\nConventional System:\n');
fprintf('Average SINR : %.2f dB\n',mean(SINR_conv_dB));
fprintf('Spectrum Efficiency : %.3f bits/s/Hz\n',avgSE_conv);
fprintf('Throughput : %.2f Mbps\n',throughput_conv/1e6);
fprintf('Average Power : %.3f W\n',avgPower_conv);

fprintf('\nProposed System:\n');
fprintf('Average SINR : %.2f dB\n',mean(SINR_prop_dB));
fprintf('Spectrum Efficiency : %.3f bits/s/Hz\n',avgSE_prop);
fprintf('Throughput : %.2f Mbps\n',throughput_prop/1e6);
fprintf('Average Power : %.3f W\n',avgPower_prop);

fprintf('\nSpectrum Efficiency Improvement = %.2f %%\n', ...
    SE_improvement);

%% Output Graph 1

figure;
scatter(BS(:,1),BS(:,2),100,'filled');
```

```
hold on;
scatter(users(:,1),users(:,2),20,'filled');

xlabel('X Position (m)');
ylabel('Y Position (m)');
title('Dense Communication Zone');
legend('Base Stations','Users');
grid on;

%% Output Graph 2

figure;
plot(1:numUsers,SINR_conv_dB,'LineWidth',1.5);
hold on;
plot(1:numUsers,SINR_prop_dB,'LineWidth',1.5);

xlabel('User Number');
ylabel('SINR (dB)');
title('SINR Comparison');
legend('Conventional','Proposed');
grid on;

%% Output Graph 3

figure;
bar([avgSE_conv avgSE_prop]);

set(gca,'XTickLabel',{'Conventional','Proposed'});
ylabel('Spectrum Efficiency (bits/s/Hz)');
title('Spectrum Efficiency Comparison');
grid on;

%% Output Graph 4
figure;
bar([avgPower_conv avgPower_prop]);
```

```
set(gca,'XTickLabel',{'Conventional','Proposed'});
ylabel('Average Transmit Power (W)');
title('Transmit Power Comparison');
grid on;
```

Output:

The MATLAB program produces:

1. Dense Communication Zone Graph – displays the locations of base stations and users.
2. SINR Comparison Graph – compares conventional and proposed systems.
3. Spectrum Efficiency Comparison Graph – shows improvement in bits/s/Hz.
4. Transmit Power Comparison Graph – compares the power used by both systems.